JOINT ENTRANCE EXAMINATION (MAIN) JULY -2022
(CANDIDATE RESPONSE SHEET)

Paper/Subject	B.E/B.Tech.(Paper 1)
Exam Date	27 Jul 2022
Exam Slot	2

Question ID:100401
Topic Name: Mathematics-Section A
Question:

The domain of the function $f(x) = \sin^{-1}[2x^2 - 3] + \log_2\left(\log_{\frac{1}{2}}\left(x^2 - 5x + 5\right)\right)$, where [t] is the greatest integer function, is :

A $\left(-\sqrt{\frac{5}{2}}, \frac{5-\sqrt{5}}{2}\right)$

B $\left(\frac{5-\sqrt{5}}{2}, \frac{5+\sqrt{5}}{2}\right)$

C $\left(1, \frac{5-\sqrt{5}}{2}\right)$

D $\left[1, \frac{5+\sqrt{5}}{2}\right)$

Answer Given By Candidate:**C**

Question ID:100402
Topic Name: Mathematics-Section A
Question:

Let S be the set of all (α, β), $\pi < \alpha, \beta < 2\pi$, for which the complex number $\frac{1 - i\sin\alpha}{1 + 2i\sin\alpha}$ is purely imaginary and $\frac{1 + i\cos\beta}{1 - 2i\cos\beta}$ is purely real. Let $Z_{\alpha\beta} = \sin 2\alpha + i\cos 2\beta$, $(\alpha, \beta) \in S$. Then $\sum_{(\alpha,\beta)\in S}\left(i\,Z_{\alpha\beta} + \frac{1}{i\,\overline{Z}_{\alpha\beta}}\right)$ is equal to :

A 3

B $3i$

C 1

D $2-i$

Answer Given By Candidate:**Not Attempted**

Question ID:100403
Topic Name: Mathematics-Section A

Question:

If α, β are the roots of the equation

$$x^2 - \left(5 + 3^{\sqrt{\log_3 5}} - 5^{\sqrt{\log_5 3}}\right) + 3\left(3^{(\log_3 5)^{\frac{1}{3}}} - 5^{(\log_5 3)^{\frac{2}{3}}} - 1\right) = 0,$$

then the equation, whose roots are $\alpha + \frac{1}{\beta}$ and $\beta + \frac{1}{\alpha}$, is :

A $3x^2 - 20x - 12 = 0$

B $3x^2 - 10x - 4 = 0$

C $3x^2 - 10x + 2 = 0$

D $3x^2 - 20x + 16 = 0$

Answer Given By Candidate:**Not Attempted**

Question ID:100404
Topic Name: Mathematics-Section A

Question:

Let $A = \begin{pmatrix} 4 & -2 \\ \alpha & \beta \end{pmatrix}$.

If $A^2 + \gamma A + 18I = O$, then det (A) is equal to _____.

A -18

B 18

C -50

D 50

Answer Given By Candidate:**Not Attempted**

Question ID:100405
Topic Name: Mathematics-Section A

Question:

If for $p \neq q \neq 0$, the function $f(x) = \frac{\sqrt[7]{p(729 + x)} - 3}{\sqrt[3]{729 + qx} - 9}$ is continuous at $x = 0$, then :

A $7pq\, f(0) - 1 = 0$

B $63q\,f(0)-p^2=0$

C $21q\,f(0)-p^2=0$

D $7pq\,f(0)-9=0$

Answer Given By Candidate:**Not Attempted**

Question ID:100406
Topic Name: Mathematics-Section A

Let $f(x)=2+|x|-|x-1|+|x+1|,\ x \in \mathbf{R}$.
Consider

$$(S1):\ f'\left(-\frac{3}{2}\right)+f'\left(-\frac{1}{2}\right)+f'\left(\frac{1}{2}\right)+f'\left(\frac{3}{2}\right)=2$$

$$(S2):\ \int_{-2}^{2} f(x)\,dx = 12$$

Question: Then,

A both (S1) and (S2) are correct

B both (S1) and (S2) are wrong

C only (S1) is correct

D only (S2) is correct

Answer Given By Candidate:**Not Attempted**

Question ID:100407
Topic Name: Mathematics-Section A
Question:

Let the sum of an infinite G.P., whose first term is a and the common ratio is r, be 5. Let the sum of its first five terms be $\frac{98}{25}$. Then the sum of the first 21 terms of an AP, whose first term is 10ar, n^{th} term is a_n and the common difference is $10ar^2$, is equal to :

A 21 a_{11}

B 22 a_{11}

C 15 a_{16}

D 14 a_{16}

Answer Given By Candidate:**Not Attempted**

Question ID:100408
Topic Name: Mathematics-Section A
Question: The area of the region enclosed by $y \leq 4x^2$, $x^2 \leq 9y$ and $y \leq 4$, is equal to :

A $\frac{40}{3}$

B $\frac{56}{3}$

C $\frac{112}{3}$

D $\frac{80}{3}$

Answer Given By Candidate:**Not Attempted**

Question ID:100409
Topic Name: Mathematics-Section A
Question:

$\int_0^2 \left(\left| 2x^2 - 3x \right| + \left[x - \frac{1}{2} \right] \right) dx$, where [t] is the greatest integer function, is equal to :

A $\frac{7}{6}$

B $\frac{19}{12}$

C $\frac{31}{12}$

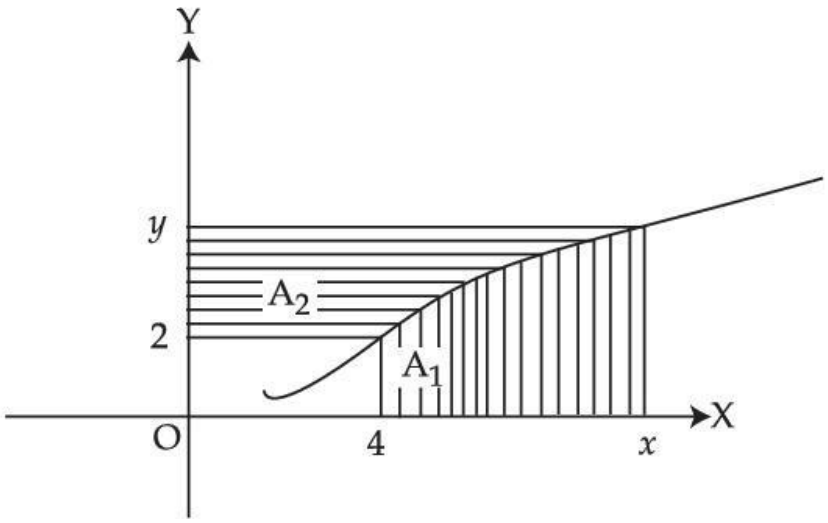
D $\frac{3}{2}$

Answer Given By Candidate:**B**

Question ID:100410
Topic Name: Mathematics-Section A

Question:

Consider a curve $y = y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x - 12y = 15$ does **NOT** pass through the point.



A $(6, 21)$

B $(8, 9)$

C $(10, -4)$

D $(12, -15)$

Answer Given By Candidate:**Not Attempted**

Question ID:100411
Topic Name: Mathematics-Section A
Question:

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 39$ and $x - y = 3$ respectively and P(2, 3) is its circumcentre. Then which of the following is **NOT** true ?

A $(AC)^2 = 9p$

B $(AC)^2 + p^2 = 136$

C $32 < \text{area } (\Delta ABC) < 36$

D $34 < \text{area } (\Delta ABC) < 38$

Answer Given By Candidate:**Not Attempted**

Question ID:100412
Topic Name: Mathematics-Section A

Question:

A circle C_1 passes through the origin O and has diameter 4 on the positive x-axis. The line $y = 2x$ gives a chord OA of circle C_1. Let C_2 be the circle with OA as a diameter. If the tangent to C_2 at the point A meets the x-axis at P and y-axis at Q, then QA : AP is equal to :

A $1 : 4$

B $1 : 5$

C $2 : 5$

D $1 : 3$

Answer Given By Candidate:**Not Attempted**

Question ID:100413
Topic Name: Mathematics-Section A
Question:

If the length of the latus rectum of a parabola, whose focus is (a, a) and the tangent at its vertex is $x + y = a$, is 16, then $|a|$ is equal to :

A $2\sqrt{2}$

B $2\sqrt{3}$

C $4\sqrt{2}$

D 4

Answer Given By Candidate:**Not Attempted**

Question ID:100414
Topic Name: Mathematics-Section A
Question:

If the length of the perpendicular drawn from the point P(a, 4, 2), $a > 0$ on the line $\frac{x+1}{2} = \frac{y-3}{3} = \frac{z-1}{-1}$ is $2\sqrt{6}$ units and $Q(\alpha_1, \alpha_2, \alpha_3)$ is the image of the point P in this line, then $a + \sum_{i=1}^{3} \alpha_i$ is equal to :

A 7

B 8

C 12

D 14

Answer Given By Candidate:**Not Attempted**

Question ID:100415
Topic Name: Mathematics-Section A
Question:

If the line of intersection of the planes $ax+by=3$ and $ax+by+cz=0$, a > 0 makes an angle $30°$ with the plane $y-z+2=0$, then the direction cosines of the line are :

A $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0$

B $\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}, 0$

C $\frac{1}{\sqrt{5}}, -\frac{2}{\sqrt{5}}, 0$

D $\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0$

Answer Given By Candidate:**Not Attempted**

Question ID:100416
Topic Name: Mathematics-Section A
Question:

Let X have a binomial distribution B(n, p) such that the sum and the product of the mean and variance of X are 24 and 128 respectively. If $P(X > n-3) = \frac{k}{2^n}$, then k is equal to :

A 528

B 529

C 629

D 630

Answer Given By Candidate:**Not Attempted**

Question ID:100417
Topic Name: Mathematics-Section A
Question:

A six faced die is biased such that

$3 \times P(\text{a prime number}) = 6 \times P(\text{a composite number}) = 2 \times P(1)$.

Let X be a random variable that counts the number of times one gets a perfect square on some throws of this die. If the die is thrown twice, then the mean of X is :

A

$\frac{3}{11}$

B $\frac{5}{11}$

C $\frac{7}{11}$

D $\frac{8}{11}$

Answer Given By Candidate:**Not Attempted**

Question ID:100418
Topic Name: Mathematics-Section A
Question:

The angle of elevation of the top P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45°. Let R be a point on AQ and from a point B, vertically above R, the angle of elevation of P is 60°. If $\angle BAQ = 30°$, $AB = d$ and the area of the trapezium PQRB is α, then the ordered pair (d, α) is :

A $\left(10(\sqrt{3}-1), 25\right)$

B $\left(10(\sqrt{3}-1), \frac{25}{2}\right)$

C $\left(10(\sqrt{3}+1), 25\right)$

D $\left(10(\sqrt{3}+1), \frac{25}{2}\right)$

Answer Given By Candidate:**Not Attempted**

Question ID:100419
Topic Name: Mathematics-Section A
Question:

Let $S = \left\{\theta \in \left(0, \frac{\pi}{2}\right) : \sum_{m=1}^{9} \sec\left(\theta + (m-1)\frac{\pi}{6}\right)\sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}}\right\}$. Then

A $S = \left\{\frac{\pi}{12}\right\}$

B $S = \left\{\frac{2\pi}{3}\right\}$

C $\sum_{\theta \in S} \theta = \frac{\pi}{2}$

D $\sum_{\theta \in S} \theta = \frac{3\pi}{4}$

Answer Given By Candidate:**Not Attempted**

Question ID:100420
Topic Name: Mathematics-Section A
Question:

If the truth value of the statement $(P \wedge (\sim R)) \rightarrow ((\sim R) \wedge Q)$ is F, then the truth value of which of the following is F ?

A $P \vee Q \rightarrow \sim R$

B $R \vee Q \rightarrow \sim P$

C $\sim(P \vee Q) \rightarrow \sim R$

D $\sim(R \vee Q) \rightarrow \sim P$

Answer Given By Candidate:**Not Attempted**

Question ID:100421
Topic Name: Mathematics-Section B
Question:

Consider a matrix $A = \begin{bmatrix} \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \\ \beta+\gamma & \gamma+\alpha & \alpha+\beta \end{bmatrix}$, where α, β, γ are three distinct natural numbers.

If $\frac{\det\left(\text{adj}\left(\text{adj}\left(\text{adj}\left(\text{adj } A\right)\right)\right)\right)}{(\alpha-\beta)^{16}(\beta-\gamma)^{16}(\gamma-\alpha)^{16}} = 2^{32} \times 3^{16}$, then the number of such 3 - tuples (α, β, γ) is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100422
Topic Name: Mathematics-Section B
Question:

The number of functions f, from the set $A = \{x \in \mathbf{N} : x^2 - 10x + 9 \leq 0\}$ to the set $B = \{n^2 : n \in \mathbf{N}\}$ such that $f(x) \leq (x-3)^2 + 1$, for every $x \in A$, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100423
Topic Name: Mathematics-Section B
Question:

Let for the 9th term in the binomial expansion of $(3+6x)^n$, in the increasing powers of $6x$, to be the greatest for $x = \frac{3}{2}$, the least value of n is n_0. If k is the ratio of the coefficient of x^6 to the coefficient of x^3, then $k+n_0$ is equal to :

Answer Given By Candidate:**Not Attempted**

Question ID:100424
Topic Name: Mathematics-Section B
Question:

$$\frac{2^3-1^3}{1\times 7}+\frac{4^3-3^3+2^3-1^3}{2\times 11}+\frac{6^3-5^3+4^3-3^3+2^3-1^3}{3\times 15}+\cdots+\frac{30^3-29^3+28^3-27^3+\ldots+2^3-1^3}{15\times 63}$$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100425
Topic Name: Mathematics-Section B
Question:

A water tank has the shape of a right circular cone with axis vertical and vertex downwards. Its semi-vertical angle is $\tan^{-1}\frac{3}{4}$. Water is poured in it at a constant rate of 6 cubic meter per hour. The rate (in square meter per hour), at which the wet curved surface area of the tank is increasing, when the depth of water in the tank is 4 meters, is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100426
Topic Name: Mathematics-Section B
Question:

For the curve C : $(x^2+y^2-3)+(x^2-y^2-1)^5=0$, the value of $3y'-y^3y''$, at the point (α, α), $\alpha > 0$, on C, is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100427
Topic Name: Mathematics-Section B

Question:

Let $f(x) = \min\{[x-1], [x-2], \ldots, [x-10]\}$

where [t] denotes the greatest integer $\le$ t. Then $\int_0^{10} f(x)\,dx + \int_0^{10} (f(x))^2\,dx + \int_0^{10} |f(x)|\,dx$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100428
Topic Name: Mathematics-Section B
Question:

Let f be a differentiable function satisfying $f(x) = \frac{2}{\sqrt{3}} \int_0^{\sqrt{3}} f\left(\frac{\lambda^2 x}{3}\right) d\lambda$, $x > 0$ and $f(1) = \sqrt{3}$. If $y = f(x)$ passes through the point $(\alpha, 6)$, then α is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100429
Topic Name: Mathematics-Section B
Question:

A common tangent T to the curves $C_1 : \frac{x^2}{4} + \frac{y^2}{9} = 1$ and $C_2 : \frac{x^2}{42} - \frac{y^2}{143} = 1$ does not pass through the fourth quadrant. If T touches C_1 at (x_1, y_1) and C_2 at (x_2, y_2), then $|2x_1 + x_2|$ is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100430
Topic Name: Mathematics-Section B
Question:

Let $\vec{a}, \vec{b}, \vec{c}$ be three non-coplanar vectors such that $\vec{a} \times \vec{b} = 4\vec{c}$, $\vec{b} \times \vec{c} = 9\vec{a}$ and $\vec{c} \times \vec{a} = \alpha\vec{b}$, $\alpha > 0$. If $|\vec{a}| + |\vec{b}| + |\vec{c}| = \frac{1}{36}$, then α is equal to _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100431
Topic Name: Physics-Section A

Question:

An expression of energy density is given by $u = \frac{\alpha}{\beta}\sin\left(\frac{\alpha x}{kt}\right)$, where α, β are constants, x is displacement, k is Boltzmann constant and t is the temperature. The dimensions of β will be :

A $[ML^2T^{-2}\theta^{-1}]$

B $[M^0L^2T^{-2}]$

C $[M^0L^0T^0]$

D $[M^0L^2T^0]$

Answer Given By Candidate:**Not Attempted**

Question ID:100432
Topic Name: Physics-Section A
Question:

A body of mass 10 kg is projected at an angle of 45° with the horizontal. The trajectory of the body is observed to pass through a point (20, 10). If T is the time of flight, then its momentum vector, at time $t = \frac{T}{\sqrt{2}}$, is _____.

[Take $g = 10\ m/s^2$]

A $100\hat{i} + (100\sqrt{2} - 200)\hat{j}$

B $100\sqrt{2}\,\hat{i} + (100 - 200\sqrt{2})\hat{j}$

C $100\hat{i} + (100 - 200\sqrt{2})\hat{j}$

D $100\sqrt{2}\,\hat{i} + (100\sqrt{2} - 200)\hat{j}$

Answer Given By Candidate:**Not Attempted**

Question ID:100433
Topic Name: Physics-Section A
Question:

A block of mass M slides down on a rough inclined plane with constant velocity. The angle made by the incline plane with horizontal is θ . The magnitude of the contact force will be :

A Mg

B Mg cosθ

C $\sqrt{Mg\sin\theta + Mg\cos\theta}$

D $Mg\sin\theta\sqrt{1+\mu}$

Answer Given By Candidate:**D**

Question ID:100434
Topic Name: Physics-Section A
Question:

A block 'A' takes 2 s to slide down a frictionless incline of 30° and length 'l', kept inside a lift going up with uniform velocity 'v'. If the incline is changed to 45°, the time taken by the block, to slide down the incline, will be approximately :

A 2.66 s

B 0.83 s

C 1.68 s

D 0.70 s

Answer Given By Candidate:**A**

Question ID:100435
Topic Name: Physics-Section A
Question:

The velocity of the bullet becomes one third after it penetrates 4 cm in a wooden block. Assuming that bullet is facing a constant resistance during its motion in the block. The bullet stops completely after travelling at $(4+x)$ cm inside the block. The value of x is :

A 2.0

B 1.0

C 0.5

D 1.5

Answer Given By Candidate:**A**

Question ID:100436
Topic Name: Physics-Section A
Question:

A body of mass m is projected with velocity λv_e in vertically upward direction from the surface of the earth into space. It is given that v_e is escape velocity and $\lambda < 1$. If air resistance is considered to be negligible, then the maximum height from the centre of earth, to which the body can go, will be :

(R : radius of earth)

A

$\frac{R}{1+\lambda^2}$

B $\frac{R}{1-\lambda^2}$

C $\frac{R}{1-\lambda}$

D $\frac{\lambda^2 R}{1-\lambda^2}$

Answer Given By Candidate:**B**

Question ID:100437
Topic Name: Physics-Section A
Question:

A steel wire of length 3.2 m ($Y_s = 2.0 \times 10^{11}$ Nm^{-2}) and a copper wire of length 4.4 m ($Y_c = 1.1 \times 10^{11}$ Nm^{-2}), both of radius 1.4 mm are connected end to end. When stretched by a load, the net elongation is found to be 1.4 mm. The load applied, in Newton, will be : (Given $\pi = \frac{22}{7}$)

A 360

B 180

C 1080

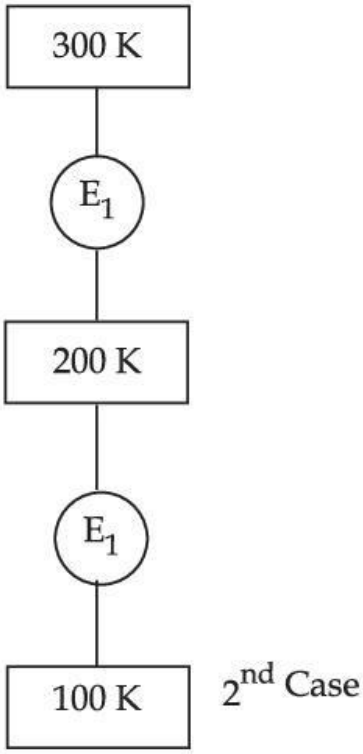
D 154

Answer Given By Candidate:**Not Attempted**

Question ID:100438
Topic Name: Physics-Section A

Question:

In 1st case, Carnot engine operates between temperatures 300 K and 100 K. In 2nd case, as shown in the figure, a combination of two engines is used. The efficiency of this combination (in 2nd case) will be :



2nd Case

A same as the 1st case.

B always greater than the 1st case.

C always less than the 1st case.

D may increase or decrease with respect to the 1st case.

Answer Given By Candidate: **A**

Question ID: 100439
Topic Name: Physics-Section A
Question:

Which statements are correct about degrees of freedom ?

(A) A molecule with n degrees of freedom has n^2 different ways of storing energy.

(B) Each degree of freedom is associated with $\frac{1}{2}$ RT average energy per mole.

(C) A monatomic gas molecule has 1 rotational degree of freedom where as diatomic molecule has 2 rotational degrees of freedom.

(D) CH_4 has a total of 6 degrees of freedom.

Choose the **correct** answer from the options given below :

A (B) and (C) only

B (B) and (D) only

C (A) and (B) only

D (C) and (D) only

Answer Given By Candidate:**B**

Question ID:100440
Topic Name: Physics-Section A
Question:

A charge of 4 μC is to be divided into two. The distance between the two divided charges is constant. The magnitude of the divided charges so that the force between them is maximum, will be :

A 1 μC and 3 μC

B 2 μC and 2 μC

C 0 and 4 μC

D 1.5 μC and 2.5 μC

Answer Given By Candidate:**C**

Question ID:100441
Topic Name: Physics-Section A
Question:

(A) The drift velocity of electrons decreases with the increase in the temperature of conductor.
(B) The drift velocity is inversely proportional to the area of cross-section of given conductor.
(C) The drift velocity does not depend on the applied potential difference to the conductor.
(D) The drift velocity of electron is inversely proportional to the length of the conductor.
(E) The drift velocity increases with the increase in the temperature of conductor.

Choose the **correct** answer from the options given below :

A (A) and (B) only

B (A) and (D) only

C (B) and (E) only **D**

(B) and (C) only

Answer Given By Candidate:**B**

Question ID:100442
Topic Name: Physics-Section A
Question:

A compass needle of oscillation magnetometer oscillates 20 times per minute at a place P of dip 30°. The number of oscillations per minute become 10 at another place Q of 60° dip. The ratio of the total magnetic field at the two places ($B_Q : B_P$) is :

A $\sqrt{3} : 4$

B $4 : \sqrt{3}$

C $\sqrt{3} : 2$

D $2 : \sqrt{3}$

Answer Given By Candidate:**B**

Question ID:100443
Topic Name: Physics-Section A
Question:

A cyclotron is used to accelerate protons. If the operating magnetic field is 1.0 T and the radius of the cyclotron 'dees' is 60 cm, the kinetic energy of the accelerated protons in MeV will be :

[use $m_p = 1.6 \times 10^{-27}$ kg, $e = 1.6 \times 10^{-19}$ C]

A 12

B 18

C 16

D 32

Answer Given By Candidate:**B**

Question ID:100444
Topic Name: Physics-Section A
Question:

A series LCR circuit has L = 0.01 H, R = 10 Ω and C = 1μF and it is connected to ac voltage of amplitude (V_m) 50 V. At frequency 60% lower than resonant frequency, the amplitude of current will be approximately :

A 466 mA

B 312 mA

C 238 mA

D 196 mA

Answer Given By Candidate:**C**

Question ID:100445
Topic Name: Physics-Section A
Question:

Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.

(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.

(C) Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.

(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

(E) The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the **most appropriate** answer from the options given below :

A (D) only

B (B) and (D) only

C (B), (C) and (E) only

D (A), (B) and (E) only

Answer Given By Candidate:**B**

Question ID:100446
Topic Name: Physics-Section A
Question:

Two coherent sources of light interfere. The intensity ratio of two sources is 1 : 4. For this interference pattern if the value of $\frac{I_{max} + I_{min}}{I_{max} - I_{min}}$ is equal to $\frac{2\alpha + 1}{\beta + 3}$, then $\frac{\alpha}{\beta}$ will be :

A 1.5

B 2

C 0.5

D 1

Answer Given By Candidate:**Not Attempted**

Question ID:100447
Topic Name: Physics-Section A
Question:

With reference to the observations in photo-electric effect, identify the correct statements from below :

(A) The square of maximum velocity of photoelectrons varies linearly with frequency of incident light.

(B) The value of saturation current increases on moving the source of light away from the metal surface.

(C) The maximum kinetic energy of photo-electrons decreases on decreasing the power of LED (light emitting diode) source of light.

(D) The immediate emission of photo-electrons out of metal surface can not be explained by particle nature of light/electromagnetic waves.

(E) Existence of threshold wavelength can not be explained by wave nature of light/ electromagnetic waves.

Choose the **correct** answer from the options given below :

A (A) and (B) only

B (A) and (E) only

C (C) and (E) only

D (D) and (E) only

Answer Given By Candidate:**B**

Question ID:100448
Topic Name: Physics-Section A
Question:

The activity of a radioactive material is 6.4×10^{-4} curie. Its half life is 5 days. The activity will become 5×10^{-6} curie after :

A 7 days

B 15 days

C 25 days

D 35 days

Answer Given By Candidate:**D**

Question ID:100449
Topic Name: Physics-Section A

Question:

For a constant collector-emitter voltage of 8 V, the collector current of a transistor reached to the value of 6 mA from 4 mA, whereas base current changed from 20 μA to 25 μA value. If transistor is in active state, small signal current gain (current amplification factor) will be :

A 240

B 400

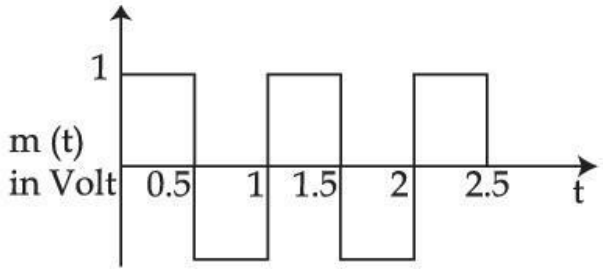
C 0.0025

D 200

Answer Given By Candidate:**B**

Question ID:100450
Topic Name: Physics-Section A
Question:

A square wave of the modulating signal is shown in the figure. The carrier wave is given by $C(t) = 5 \sin(8 \pi t)$ Volt. The modulation index is :



A 0.2

B 0.1

C 0.3

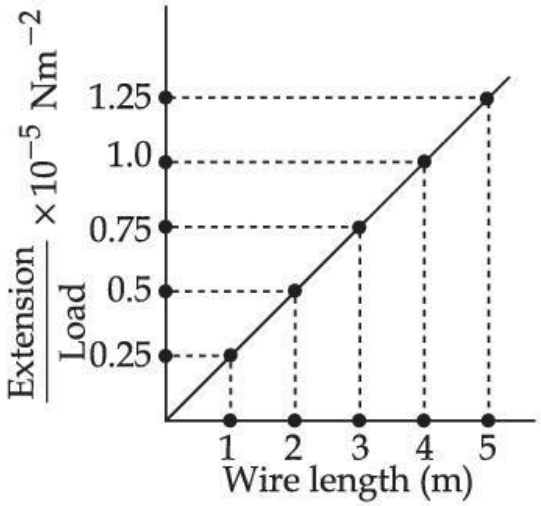
D 0.4

Answer Given By Candidate:**A**

Question ID:100451
Topic Name: Physics-Section B

Question:

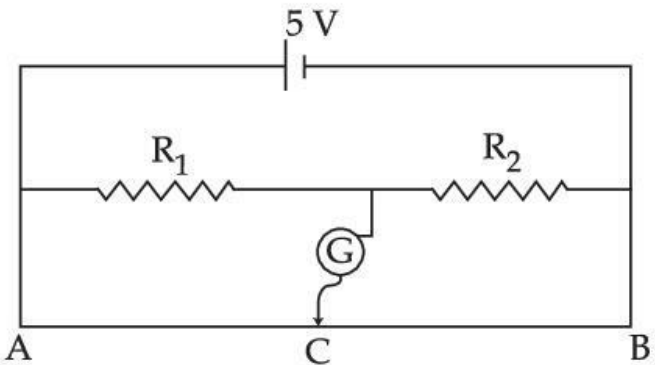
In an experiment to determine the Young's modulus, steel wires of five different lengths (1, 2, 3, 4, and 5 m) but of same cross section (2 mm^2) were taken and curves between extenstion and load were obtained. The slope (extension/load) of the curves were plotted with the wire length and the following graph is obtained. If the Young's modulus of given steel wires is $x \times 10^{11}$ Nm^{-2}, then the value of x is _____.



Answer Given By Candidate:**Not Attempted**

Question ID:100452
Topic Name: Physics-Section B
Question:

In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be _____ cm.

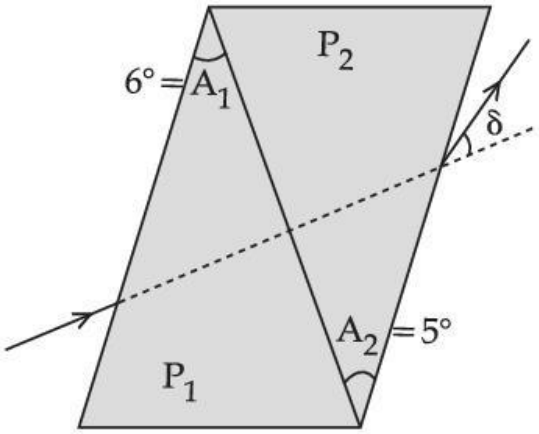


Answer Given By Candidate:**Not Attempted**

Question ID:100453
Topic Name: Physics-Section B

Question:

A thin prism of angle 6° and refractive index for yellow light (n_Y)1.5 is combined with another prism of angle 5° and $n_Y = 1.55$. The combination produces no dispersion. The net average deviation (δ) produced by the combination is $\left(\frac{1}{x}\right)^\circ$. The value of x is _____ .



Answer Given By Candidate:**Not Attempted**

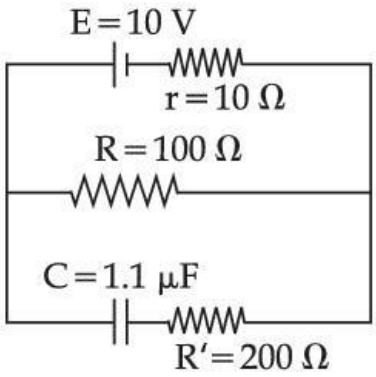
Question ID:100454
Topic Name: Physics-Section B
Question:

A conducting circular loop is placed in X - Y plane in presence of magnetic field $\vec{B} = \left(3t^3 \hat{j} + 3t^2 \hat{k}\right)$ in SI unit. If the radius of the loop is 1 m, the induced emf in the loop, at time, t = 2 s is nπ V. The value of n is _____.

Answer Given By Candidate:**48**

Question ID:100455
Topic Name: Physics-Section B
Question:

As show in the figure, in steady state, the charge stored in the capacitor is _____ $\times 10^{-6}$ C.



Answer Given By Candidate:**1**

Question ID:100456
Topic Name: Physics-Section B
Question:

A parallel plate capacitor with width 4 cm, length 8 cm and separation between the plates of 4 mm is connected to a battery of 20 V. A dielectric slab of dielectric constant 5 having length 1 cm, width 4 cm and thickness 4 mm is inserted between the plates of parallel plate capacitor. The electrostatic energy of this system will be _____ ϵ_0 J. (Where ϵ_0 is the permittivity of free space)

Answer Given By Candidate:**Not Attempted**

Question ID:100457
Topic Name: Physics-Section B
Question:

A wire of length 30 cm, stretched between rigid supports, has it's n^{th} and $(n+1)^{th}$ harmonics at 400 Hz and 450 Hz, respectively. If tension in the string is 2700 N, it's linear mass density is _____ kg/m.

Answer Given By Candidate:**3**

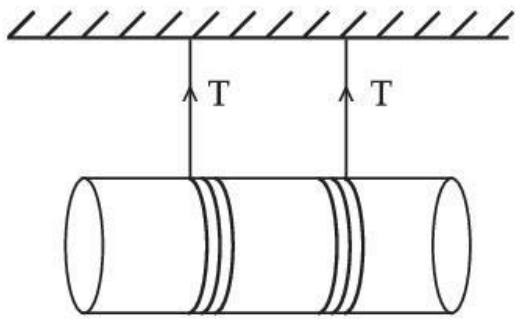
Question ID:100458
Topic Name: Physics-Section B
Question:

A spherical soap bubble of radius 3 cm is formed inside another spherical soap bubble of radius 6 cm. If the internal pressure of the smaller bubble of radius 3 cm in the above system is equal to the internal pressure of the another single soap bubble of radius r cm. The value of r is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100459
Topic Name: Physics-Section B
Question:

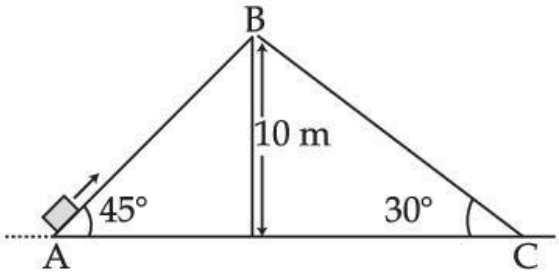
A solid cylinder length is suspended symmetrically through two massless strings, as shown in the figure. The distance from the initial rest position, the cylinder should by unbinding the strings to achieve a speed of 4 ms^{-1}, is _____ cm. (take $g = 10\ ms^{-2}$)



Answer Given By Candidate:**Not Attempted**

Question ID:100460
Topic Name: Physics-Section B
Question:

Two inclined planes are placed as shown in figure. A block is projected from the Point A of inclined plane AB along its surface with a velocity just sufficient to carry it to the top Point B at a height 10 m. After reaching the Point B the block slides down on inclined plane BC. Time it takes to reach to the point C from point A is $t(\sqrt{2}+1)$ s. The value of t is _____.
(use $g = 10\ m/s^2$)



Answer Given By Candidate:**Not Attempted**

Question ID:100461
Topic Name: Chemistry-Section A
Question:

The correct decreasing order of energy for the orbitals having, following set of quantum numbers :

(A) $n=3, l=0, m=0$
(B) $n=4, l=0, m=0$
(C) $n=3, l=1, m=0$
(D) $n=3, l=2, m=1$

is :

A (D) > (B) > (C) > (A)

B (B) > (D) > (C) > (A)

C (C) > (B) > (D) > (A)

D (B) > (C) > (D) > (A)

Answer Given By Candidate:**A**

Question ID:100462
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

	List - I		**List - II**
(A)	$\psi_{MO}=\psi_A-\psi_B$	(I)	Dipole moment
(B)	$\mu=Q\times r$	(II)	Bonding molecular orbital
(C)	$\frac{N_b-N_a}{2}$	(III)	Anti-bonding molecular orbital
(D)	$\psi_{MO}=\psi_A+\psi_B$	(IV)	Bond order

Choose the **correct** answer from the options given below :

Question:

A (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

B (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

C (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

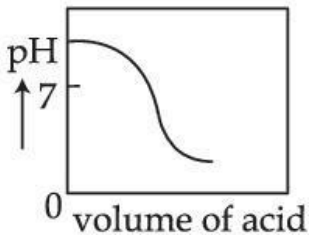
D (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Answer Given By Candidate:**C**

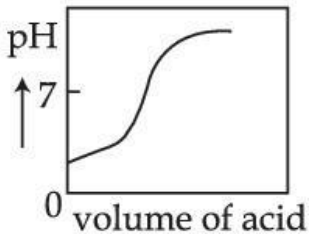
Question ID:100463
Topic Name: Chemistry-Section A
Question:

The plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like :

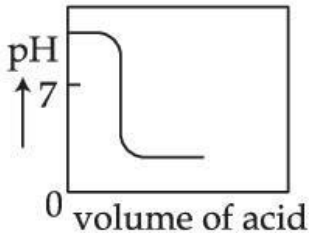
A



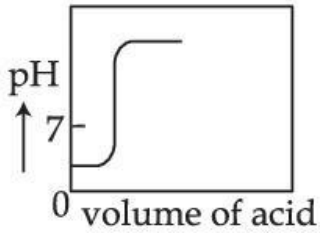
B



C



D



Answer Given By Candidate:**C**

Question ID:100464
Topic Name: Chemistry-Section A
Question:

Given below are two statements :

Statement I : For KI, molar conductivity increases steeply with dilution

Statement II : For carbonic acid, molar conductivity increases slowly with dilution

In the light of the above statements, choose the **correct** answer from the options given below :

A Both **Statement I** and **Statement II** are true

B Both **Statement I** and **Statement II** are false

C **Statement I** is true but **Statement II** is false

D **Statement I** is false but **Statement II** is true

Answer Given By Candidate:**A**

Question ID:100465
Topic Name: Chemistry-Section A
Question:

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Dissolved substances can be removed from a colloidal solution by diffusion through a parchment paper.

Reason (R) : Particles in a true solution cannot pass through parchment paper but the colloidal particles can pass through the parchment paper.

In the light of the above statements, choose the **correct** answer from the options given below :

A Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

B Both **(A)** and **(R)** are correct but **(R)** is **not** the correct explanation of **(A)**

C **(A)** is correct but **(R)** is not correct

D **(A)** is not correct but **(R)** is correct

Answer Given By Candidate:**C**

Question ID:100466
Topic Name: Chemistry-Section A
Question:

Outermost electronic configurations of four elements A, B, C, D are given below :

(A) $3s^2$

(B) $3s^2 3p^1$

(C) $3s^2 3p^3$

(D) $3s^2 3p^4$

The **correct** order of first ionization enthalpy for them is :

A (A) < (B) < (C) < (D)

B (B) < (A) < (D) < (C)

C (B) < (D) < (A) < (C)

D (B) < (A) < (C) < (D)

Answer Given By Candidate:**B**

Question ID:100467
Topic Name: Chemistry-Section A
Question:

An element A of group 1 shows similarity to an element B belonging to group 2. If A has maximum hydration enthalpy in group 1 then B is :

A Mg

B Be

C Ca

D Sr

Answer Given By Candidate:**A**

Question ID:100468
Topic Name: Chemistry-Section A
Question:

Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Boron is unable to form BF_6^{3-} .

Reason (R) : Size of B is very small.

In the light of the above statements, choose the **correct** answer from the options given below :

A

Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

B Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

C **(A)** is true but **(R)** is false

D
(A) is false but **(R)** is true

Answer Given By Candidate:**B**

Question ID:100469
Topic Name: Chemistry-Section A

Question: In neutral or alkaline solution, MnO_4^- oxidises thiosulphate to :

A $S_2O_7^{2-}$

B $S_2O_8^{2-}$

C SO_3^{2-}

D SO_4^{2-}

Answer Given By Candidate:**Not Attempted**

Question ID:100470
Topic Name: Chemistry-Section A

Question: Low oxidation state of metals in their complexes are common when ligands :

A have good π-accepting character

B have good σ-donor character

C are having good π-donating ability

D are having poor σ-donating ability

Answer Given By Candidate:**A**

Question ID:100471
Topic Name: Chemistry-Section A

Question:

Given below are two statements :

Statement I : The non bio-degradable fly ash and slag from steel industry can be used by cement industry.

Statement II : The fuel obtained from plastic waste is lead free.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

A Both **Statement I** and **Statement II** are correct

B Both **Statement I** and **Statement II** are incorrect

C **Statement I** is correct but **Statement II** is incorrect

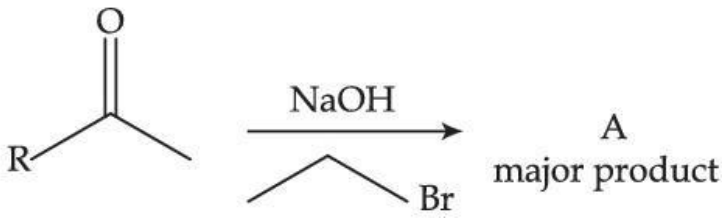
D **Statement I** is incorrect but **Statement II** is correct

Answer Given By Candidate:**A**

Question ID:100472

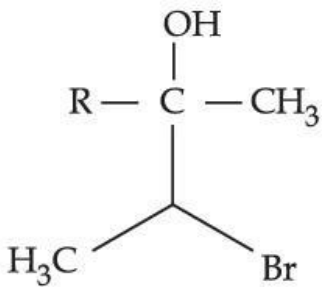
Topic Name: Chemistry-Section A

The structure of A in the given reaction is :

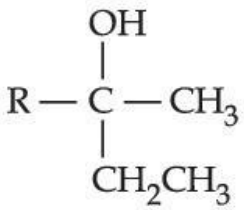


Question:

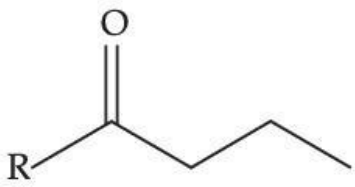
A



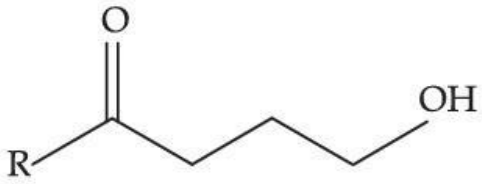
B



C



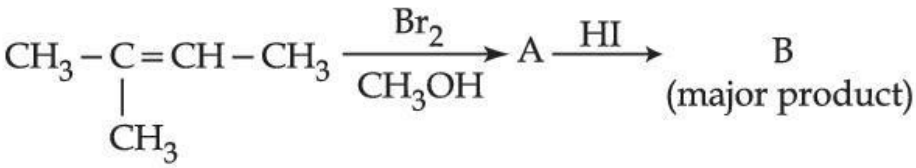
D



Answer Given By Candidate:**B**

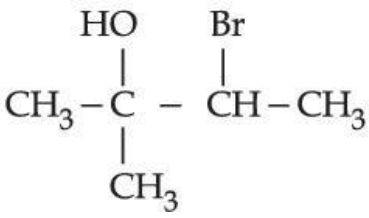
Question ID:100473
Topic Name: Chemistry-Section A

Major product 'B' of the following reaction sequence is :

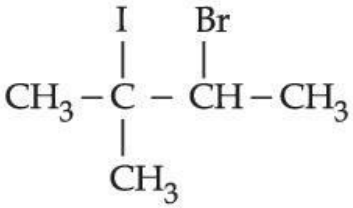


Question:

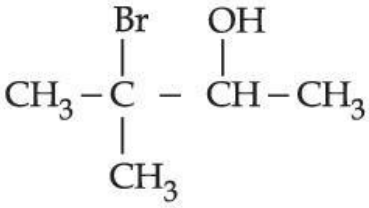
A



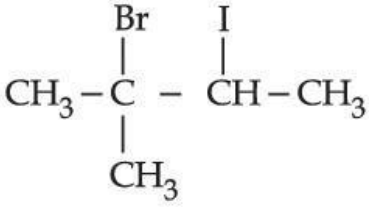
B



C



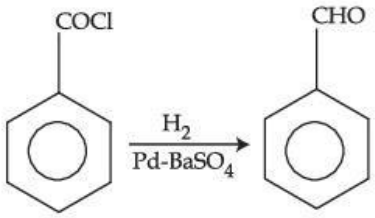
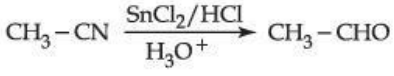
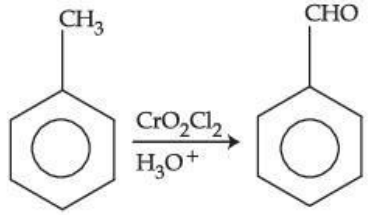
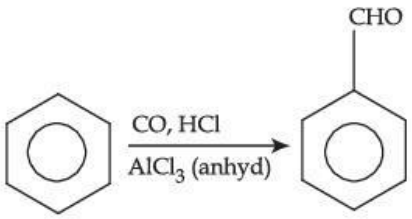
D



Answer Given By Candidate:**B**

Question ID:100474
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

List - I		**List - II**
(A)		(I) Gatterman Koch reaction
(B)		(II) Etard reaction
(C)		(III) Stephen reaction
(D)		(IV) Rosenmund reaction

Question: Choose the **correct** answer from the options given below :

A (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

B (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

C (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

D (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Answer Given By Candidate:**A**

Question ID:100475

Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

List - I (Polymer)		**List - II (Monomer)**
(A)	Neoprene	(I) Acrylonitrile
(B)	Teflon	(II) Chloroprene
(C)	Acrilan	(III) Tetrafluoroethene
(D)	Natural rubber	(IV) Isoprene

Question: Choose the **correct** answer from the options given below :

A (A) - (II), (B) - (III), (C) - (I), (D) - (IV)

B (A) - (II), (B) - (I), (C) - (III), (D) - (IV)

C (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

D

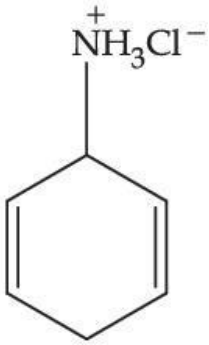
(A) - (I), (B) - (II), (C) - (III), (D) - (IV)

Answer Given By Candidate:**A**

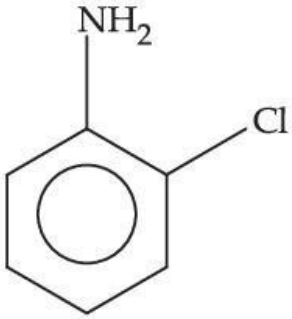
Question ID:100476
Topic Name: Chemistry-Section A
Question:

An organic compound 'A' contains nitrogen and chlorine. It dissolves readily in water to give a solution that turns litmus red. Titration of compound 'A' with standard base indicates that the molecular weight of 'A' is 131 ± 2. When a sample of 'A' is treated with aq. NaOH, a liquid separates which contains N but not Cl. Treatment of the obtained liquid with nitrous acid followed by phenol gives orange precipitate. The compound 'A' is :

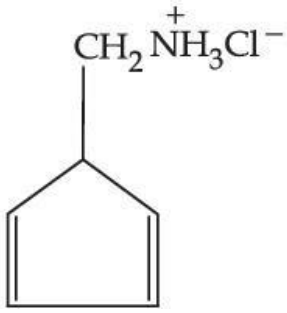
A



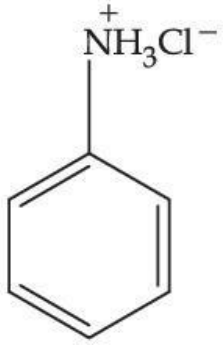
B



C



D



Answer Given By Candidate:**Not Attempted**

Question ID:100477
Topic Name: Chemistry-Section A

Question:

Match **List - I** with **List - II**.

	List - I		List - II
(A)	Glucose + HI	(I)	Gluconic acid
(B)	Glucose + Br_2 water	(II)	Glucose pentacetate
(C)	Glucose + acetic anhydride	(III)	Saccharic acid
(D)	Glucose + HNO_3	(IV)	Hexane

Choose the **correct** answer from the options given below :

A (A) - (IV), (B) - (I), (C) - (II), (D) - (III)

B (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

C (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

D (A) - (I), (B) - (III), (C) - (IV), (D) - (II)

Answer Given By Candidate:**B**

Question ID:100478
Topic Name: Chemistry-Section A

Question:

Which of the following enhances the lathering property of soap ?

A Sodium stearate

B Sodium carbonate

C Sodium rosinate

D Trisodium phosphate

Answer Given By Candidate:**B**

Question ID:100479
Topic Name: Chemistry-Section A

Match **List - I** with **List - II**.

List - I (Mixture)		**List - II (Purification Process)**
(A)	Chloroform & Aniline	(I) Steam distillation
(B)	Benzoic acid & Napthalene	(II) Sublimation
(C)	Water & Aniline	(III) Distillation
(D)	Napthalene & Sodium chloride	(IV) Crystallisation

Choose the **correct** answer from the options given below :

Question:

A (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

B (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

C (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

D (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Answer Given By Candidate:**D**

Question ID:100480
Topic Name: Chemistry-Section A
Question:

Fe^{3+} cation gives a prussian blue precipitate on addition of potassium ferrocyanide solution due to the formation of :

A $[Fe(H_2O)_6]_2\ [Fe(CN)_6]$

B $Fe_2[Fe(CN)_6]_2$

C $Fe_3[Fe(OH)_2\ (CN)_4]_2$

D $Fe_4[Fe(CN)_6]_3$

Answer Given By Candidate:**Not Attempted**

Question ID:100481
Topic Name: Chemistry-Section B
Question:

The normality of H_2SO_4 in the solution obtained on mixing 100 mL of 0.1 M H_2SO_4 with 50 mL of 0.1 M NaOH is _____ $\times 10^{-1}$ N. (Nearest Integer)

Answer Given By Candidate:**5**

Question ID:100482
Topic Name: Chemistry-Section B

Question:

For a real gas at 25°C temperature and high pressure (99 bar) the value of compressibility factor is 2, so the value of Vander Waal's constant 'b' should be _____ $\times 10^{-2}$ L mol^{-1} (Nearest integer)

(Given R = 0.083 L bar $K^{-1}mol^{-1}$)

Answer Given By Candidate:**Not Attempted**

Question ID:100483
Topic Name: Chemistry-Section B
Question:

A gas (Molar mass=280 g mol^{-1}) was burnt in excess O_2 in a constant volume calorimeter and during combustion the temperature of calorimeter increased from 298.0 K to 298.45 K. If the heat capacity of calorimeter is 2.5 kJ K^{-1} and enthalpy of combustion of gas is 9 kJ mol^{-1} then amount of gas burnt is _____ g. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100484
Topic Name: Chemistry-Section B
Question:

When a certain amount of solid A is dissolved in 100 g of water at 25°C to make a dilute solution, the vapour pressure of the solution is reduced to one-half of that of pure water. The vapour pressure of pure water is 23.76 mmHg. The number of moles of solute A added is _____. (Nearest Integer)

Answer Given By Candidate:**Not Attempted**

Question ID:100485
Topic Name: Chemistry-Section B
Question:

[A] $\rightarrow$ [B]

Reactant Product

If formation of compound [B] follows the first order of kinetics and after 70 minutes the concentration of [A] was found to be half of its initial concentration. Then the rate constant of the reaction is $x \times 10^{-6}$ s^{-1}. The value of x is _____. (Nearest Integer)

Answer Given By Candidate:**165**

Question ID:100486
Topic Name: Chemistry-Section B
Question:

Among the following ores Bauxite, Siderite, Cuprite, Calamine, Haematite, Kaolinite, Malachite, Magnetite, Sphalerite, Limonite, Cryolite, the number of principal ores if iron is _____.

Answer Given By Candidate:**4**

Question ID:100487

Topic Name: Chemistry-Section B
Question:

The oxidation state of manganese in the product obtained in a reaction of potassium permanganate and hydrogen peroxide in basic medium is _____.

Answer Given By Candidate:**Not Attempted**

Question ID:100488
Topic Name: Chemistry-Section B
Question:

The number of molecule(s) or ion(s) from the following having non-planar structure is _____.

NO_3^-, H_2O_2, BF_3, PCl_3, XeF_4, SF_4, XeO_3, PH_4^+, SO_3, $[Al(OH)_4]^-$

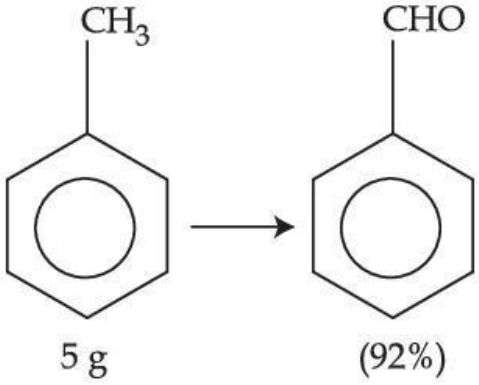
Answer Given By Candidate:**Not Attempted**

Question ID:100489
Topic Name: Chemistry-Section B
Question:

The spin only magnetic moment of the complex present in Fehling's reagent is _____ B.M. (Nearest integer).

Answer Given By Candidate:**2**

Question ID:100490
Topic Name: Chemistry-Section B
Question:



In the above reaction, 5 g of toluene is converted into benzaldehyde with 92% yield. The amount of benzaldehyde produced is _____ $\times 10^{-2}$ g. (Nearest integer)

Answer Given By Candidate:**Not Attempted**